

Hovik Grigoryan

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EDUCATION

Cornell University, College of Engineering

2023-2027

B.S. Mechanical Engineering, Minor in Aerospace Engineering (Davis Scholar)

- Relevant Courses: Finite Element Analysis, Intermediate Fluid Dynamics, Spaceflight Mechanics, Relativity & Astrophysics, Propulsion of Aircraft & Rockets, Heat Transfer, Electric Drive Vehicles, Linear Algebra, Differential Equations, System Dynamics, Statics & Mechanics of Solids, Laser & Photonics, Oscillations Waves & Quantum Physics

United World Colleges Dilijan (UWCD)

2021-2023

International Baccalaureate, Bilingual Diploma

RESEARCH EXPERIENCE

Aerospace Adversary Lab (Prof. G. Falco), Cornell University, Department of Aerospace Engineering

Jan 2026 - Present

Undergraduate Researcher

Ithaca, NY

Fiber Optic Laser Position Sensing System (ongoing)

- Codesigned a fiber sensor to localize laser incidence on a surface using a 4×4 fiber array and camera based detection
- CAD-modeled and 3D-printed all mechanical parts: detection surface, fiber grid mount, camera housing, enclosures
- Developed a MATLAB script; interpolation based position estimation; UI with the CAD model for real-time tracking

Multi-Agent Swarm Coordination via Neural Networks (ongoing, 2 months – manuscript in preparation)

- Contributing to neural network models for multi-drone cooperation, via VS code SSH remote development
- Refined multiple model architectures and hyperparameter configurations, reducing average belief model error by 70%
- Achieved consistent task success across trials, demonstrated robust coordination behavior, performed data collection

Autonomous Drone Navigation

- Assembled, 3D-printed, and repaired drone hardware, including a rebuild after a crash test, rewired drones from scratch
- Designed and calibrated a localization system using Marvelmind ultrasonic beacons, achieving 2cm position accuracy
- Integrated positioning data with flight control software (QGroundControl), achieved autonomous takeoff and landing

SIOS Lab (Prof. D. Savransky), Cornell University, Department of Aerospace Engineering

May 2025 - Present

Undergraduate Researcher (Engineering Learning Initiatives Award Recipient)

Ithaca, NY

- Contributed to NASA's Nancy Grace Roman Space Telescope project, supporting exoplanet observation and detection
- Developed Python algorithms estimating exoplanet detectability (Δmag) under integration-time constraints, made tutorials
- Built 10+ scripts and mission-level analysis tools for exoplanet yield estimation, survey design, and observation planning
- Developed scripts comparing SIOS Lab approach to flux and integration time calculations with the NASA method

Katz Lab (Prof. O. Katz), Cornell University, Department of Applied Engineering Physics

Jan 2025 - May 2025

Undergraduate Researcher

Ithaca, NY

- Supported development of a quantum computer by identifying high-precision accelerometers for the assembly platform
- Researched and compared 15 accelerometers solutions from 6 manufacturers including Brüel & Kjær, Kistler, Wilcoxon
- Coordinated with engineers from vendors and recommended accelerometers fit for the lab's operation and budget

PUBLICATIONS

Dmitry Savransky, Rifah Tasnim, Saanika Choudhary, **Hovik Grigoryan**, Justin Hom, Bijan Nemati, Neil Zimmerman, Ramya Anche, Nicholas Phillips, Louie Dones, Parth Mittal, Amira Razack, Savaas Iqbal, Javier Majumdar, Schuyler G. Wolff, and Vanessa P. Bailey, "The Roman Coronagraph Database and Tools", Proc. SPIE 14145, Space Telescopes and Instrumentation 2026: Optical, Infrared, and Millimeter Wave, 141454S (21 Aug 2026); <https://doi.org/10.1117/12.3103065>

Dmitry Savransky, Rifah Tasnim, Saanika Choudhary, **Hovik Grigoryan**, Justin Hom, Bijan Nemati, Neil Zimmerman, Ramya Anche, Nicholas Phillips, Louie Dones, Parth Mittal, Amira Razack, Savaas Iqbal, Javier Majumdar, Schuyler G. Wolff, and Vanessa P. Bailey, "The Roman Coronagraph Database and Tools," (poster), *SPIE Astronomical Telescopes + Instrumentation*, Copenhagen, Denmark, 2026.

TEACHING EXPERIENCE

System Dynamics Lab TA, Cornell University, Department of Mechanical Engineering

Aug 2026 - Present

Teaching Assistant

Ithaca, NY

- Lead two weekly lab sections (~20 students each), guiding exercises with function generators, oscilloscopes, motors
- Instruct students on Keysight BenchVue software for data acquisition and analysis for system dynamics experiments
- Support student troubleshooting of hardware setups and lab procedures, give feedback on the data, grade lab reports
- Participated in TA trainings to prepare for sections, improve presentation skills in delivering technical material

SELECTED PROJECTS

Autonomous Robot, Cornell University

March 2026 - May 2026

Class Final Project

Ithaca, NY

- Designed and built an autonomous robot (team of 3), used color sensor, QTI sensors, Arduino microcontroller, motors
- Wired the robot, used h-bridges, resistors; contributed to the code (C) and strategy design, designed and 3D printed parts
- Robot successfully was able to collect and hold 15 small cubes within 1 minute, consistently stayed within the playfield

Coffee Heater, Cornell University

Nov 2025 - Dec 2025

Class Final Project

Ithaca, NY

- Designed and constructed a coffee heater for System Dynamics class, with an implemented PID controller
- The coffee heater achieved a rise time of 144 seconds, settling time of 487.5 seconds, and a max control effort of 350 °C
- In the team, I was responsible for creating the control effort and temperature plots for the design, using MATLAB

N₂ Pulse Laser, Cornell University

Aug 2023 - Dec 2023

Class Final Project

Ithaca, NY

- Designed and built an N₂ pulse laser utilizing a Thyatron and a 555-timer as part of the Lasers and Photonics course
- Developed a pulse generator. The laser produced a 337nm wavelength light, with output laser peak power of 14.5 kW
- Selected and integrated components including diodes, transformers, and resistors to optimize laser performance

Competition Robot, FIRST Global Challenge Team Armenia

Aug 2022 - Oct 2022

Robotics Engineer

Geneva, Switzerland

- Led Team Armenia in an international FIRST Global Challenge robotics competition, consisting of 5 members
- Designed and constructed the driving system, the climbing mechanism of the robot (achieved 2 meter elevation)
- Developed control systems using Arduino to enhance robot performance in speed and precision when collecting objects

LEADERSHIP EXPERIENCE

Cornell Armenian Student Organization, Cornell University

Aug 2024 - Aug 2025

Vice President

Ithaca, NY

- Organized over 20 events for the Armenian student organization in Cornell, consisting of 50+ members
- Created and shared an event calendar with the organization members, lead weekly Armenian language speaking sessions
- Cooperated with the Cornell Dancing Club to facilitate the incorporation of Armenian traditional dances in a show

Math tutoring club, UWC Dilijan

Nov 2021 - May 2023

President

Dilijan, Armenia

- Established and organized a math tutoring club to support students in calculus and geometry
- Developed a comprehensive selection process for tutors to ensure high quality instruction, led a team of 12 math tutors
- Matched tutors with tutees to facilitate effective learning experiences for 35+ students, led individual and group tutoring

SKILLS

Programming / Software: Fusion 360, Ansys Workbench, LabVIEW, MATLAB, Python, VS Code (Remote-SSH), Git, Linux

Hardware: Wiring, soldering, 3D printing, Arduino / microcontrollers, PID controller design, sensor integration, oscilloscopes

Languages: English (fluent), Armenian (native), Russian (fluent), German (intermediate)

CAMPUS ENGAGEMENT

Cornell Makers Lab (member), Armenian Student Organization (member, VP), Cornell Fashion Collectives (runway model SP26), Intramural Soccer (team captain, member), Cornell Philosophy Club (member)